Ariff Muhammed Ahsan Husain

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Education

National University of Singapore — Singapore

Bachelor of Engineering in Computer Engineering, Honours | Minor: Data Analytics — *Aug 2024 – Present*

- **CGPA**: 4.55/5.0 **Expected Graduation**: May 2028
- **Relevant Courses**: Data Structures & Algorithms (C++), Systems Programming, Real-time Operating Systems, Digital Circuit Design, Computer Architecture, Linear Algebra, Probability & Statistics

Technical Skills

Languages: C++, Python, C, Verilog, Java, TypeScript, SQL
Frameworks: PyTorch, Scikit-learn, Pandas, NumPy, LangChain, LangGraph, React
Databases: PostgreSQL (Supabase), ChromaDB, SQLite
DevOps / Tools: GitHub Actions, Docker, Linux/Unix, Arduino, FPGA Synthesis
EDA / IC Design: Cadence Virtuoso, Synopsys Design Compiler, VCS, hspiceD, FreePDK45

Experience

Synapxe — Singapore

Software Engineering Intern, Generative AI Team — *Jun 2026 – Present*

- Working within Synapxe's Generative AI team to enhance **Tandem** and **In-SYNc**, the national HealthTech agency's enterprise GenAI platform and **RAG-based internal chatbot**, deployed across Singapore's public healthcare system.
- Implementing **agentic AI workflows, query decomposition, and prompt engineering** using **LangChain/LangGraph** and Azure SDK, and evaluating LLM response quality with tools such as **Ragas** to improve retrieval accuracy and reduce hallucination.

Projects

Arc: ML-Powered Trading Signal Engine | *C++, Python, PyTorch, Parquet/Arrow* — May 2026 – Present

- Built a data pipeline ingesting **20 years of daily OHLCV data across 58 equities, ETFs, and Bitcoin** via yfinance, with timezone normalisation and validation for data integrity across all assets.
- Engineered **14+ normalised, asset-agnostic features** (log returns, rolling volatility, Z-score mean reversion, ATR) using **Parquet** columnar storage for efficient downstream access.
- Currently developing a **C++ feature computation engine** with SIMD/AVX2 vectorisation and lock-free data structures, alongside a **PyTorch asset embedding layer** and **HMM-based regime detection** for a multi-asset signal model.

Verialgo: Bare-Metal SoC & Algorithm Engine | *Verilog, FPGA, Dual-Port Block RAM* — Oct – Nov 2025

- Designed a fully synchronous **bare-metal SoC** in Verilog from scratch with distinct datapath and control planes; implemented custom FSMs to execute **2 parallelised sorting algorithms (Merge Sort & Cocktail Sort)** simultaneously, achieving **2× throughput** over sequential execution.
- Engineered a circular "Delta Animation Buffer" using **Dual-Port Block RAM** to snapshot algorithmic state at every clock cycle; built a pixel-stream rendering engine to drive an SPI OLED display entirely in RTL.

Baymax: Dual-MCU RTOS Study Monitor | *C, FreeRTOS, ESP32-S2, MCXC444* — Mar 2026 – Apr 2026

- Implemented **FreeRTOS sensor acquisition tasks** with **EMA filtering** on the NXP MCXC444 to monitor ambient light, sound, temperature, and humidity in real time, transmitted over UART to a paired ESP32-S2.
- Integrated the **Google Gemini API** to generate personalised, AI-driven study environment reports and **real-time Telegram notifications** at the end of each session, as part of a 4-person dual-microcontroller IoT system.

Search-and-Rescue Robot ALEX | *C++, Arduino, Raspberry Pi, LiDAR* — Mar 2025 – Apr 2025

- Programmed **Arduino Mega** firmware in C++ for motor control and obstacle detection using **Timer5 interrupts** with a custom **packet-based serial protocol** for Raspberry Pi ↔ Arduino communication; built a **TLS-secured client-server system** and integrated **LiDAR-based SLAM** for real-time mapping.

Honors & Awards

UBS Global Coding Challenge | *Top 15 Finalist — Global* — Aug 2025

- Ranked **Top 15 globally** by solving **multi-constraint algorithmic optimisation** problems in **C++** across a timed competitive programming format spanning multiple elimination rounds.

WorldQuant International Quant Competition | *Gold Level* — Jun 2025

- Achieved Gold status by engineering **alphas** in Python, backtesting against 5+ years of cross-sectional equity market data to predict short-term price movements.

NUS Hack&Roll Hackathon | *Top 15 Innovation Prize (300+ teams)* — Jan 2025

- Awarded Top 15 for *Pawgress* — a **React productivity app** built end-to-end in 24 hours, recognised for creative UX design and strong technical execution.